SAT Math Mastery · Session 11

Area and Volume

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	area of composite figures · volume of prisms, cylinders, cones, spheres and pyramids · surface area · composite and carved solids · similar figures and solids (scale factors for length, area, volume) · right triangles inside solids · unit conversion · displacement and packing
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos or on a calculator, one expression per line. Where a graph is used, tap a grey dot to read an intersection.

1. HARD · DESMOS

A scale model of a section of a long building is geometrically similar to the actual structure, with a scale factor of 1 to 20 (model to actual). If the volume of the model is 3 cubic meters, what is the volume of the actual segment, in cubic meters?

(A) 8,000 **(B) 24,000 ✓**

(C) 60 (D) 1,200

1. Volume scales with the cube of the scale factor. Type `20^3`: it returns 8000.

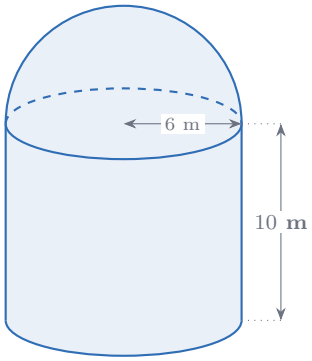
2. Type `3*20^3`: it returns 24000 cubic meters.

3. Trap check: type `3*20^2`: it returns 1200 and type `3*20`: it returns 60 come from the wrong power. Option (B).

Answer: (B)

2. HARD · DESMOS

A pavilion is designed as a cylindrical base topped by a hemispherical dome of the same radius. The cylindrical base has radius 6 meters and height 10 meters, and the dome is a hemisphere with radius 6 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 360π (B) 432π

(C) 504π ✓ (D) 576π

1. Cylinder: Type `pi*6^2*10` : it returns 1130.9734, which is 360π.

2. Hemisphere: Type `2/3*pi*6^3` : it returns 452.3893, which is 144π.

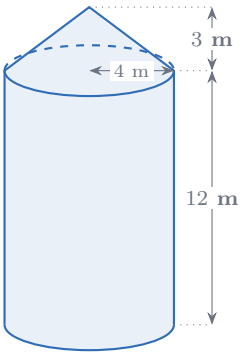
3. Total: Type `pi*6^2*10+2/3*pi*6^3` : it returns 1583.3627.

4. Match the option: type `504pi` : it returns 1583.3627, option (C).

Answer: (C)

3. HARD · DESMOS

A grain silo is designed as a cylindrical base topped by a conical roof of the same radius. The cylindrical base has radius 4 meters and height 12 meters, and the conical roof has radius 4 meters and height 3 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 208π ✓ (B) 240π

(C) 176π (D) 192π

1. Cylinder: Type `pi*4^2*12` : it returns 603.1858, which is 192π.

2. Cone: Type `1/3*pi*4^2*3` : it returns 50.2655, which is 16π.

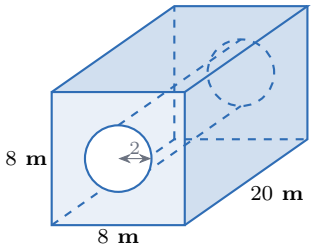
3. Total: Type `pi*4^2*12+1/3*pi*4^2*3` : it returns 653.4513.

4. Match the option: type `208pi` : it returns 653.4513, option (A).

Answer: (A)

4. Hard · Desmos

A stone block used in a restoration project is a rectangular prism measuring 8 meters by 8 meters by 20 meters, with a cylindrical channel of radius 2 meters drilled completely through its 20-meter length. What is the volume of the remaining stone, in cubic meters, in terms of π?



(A) $1280 - 160\pi$

(B) $1280 - 20\pi$

(C) $1200 - 80\pi$

(D) $1280 - 80\pi$ ✓

1. Prism: Type `8*8*20`: it returns 1280.

2. Channel (cylinder of radius 2, length 20): Type `pi*2^2*20`: it returns 251.3274, which is 80π.

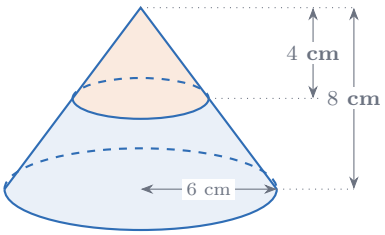
3. Remaining stone: Type `8*8*20-pi*2^2*20`: it returns 1028.6726.

4. Match the option: type `1280-80pi`: it returns 1028.6726, option (D).

Answer: (D)

5. Hard · Desmos

A cone has base radius 6 centimeters and height 8 centimeters, apex pointing up. A plane parallel to the base cuts the cone at a height of 4 centimeters from the apex, removing the smaller cone on top. What is the volume of the removed smaller cone, in cubic centimeters, in terms of π?



(A) 48π

(B) 96π

(C) 12π ✓

(D) 24π

1. Whole cone: Type `1/3*pi*6^2*8`: it returns 301.5929.

2. The small cone has height 4, half of 8, so every length is halved and the volume is multiplied by type `(4/8)^3`: it returns 0.125.

3. Small cone: Type `1/3*pi*6^2*8*(4/8)^3`: it returns 37.6991.

4. Match the option: type `12pi`: it returns 37.6991, option (C).

Answer: (C)

6. HARD · DESMOS

A cylindrical water tank has a radius of 5 meters and a height of 12 meters. What is the total surface area of the tank, including both circular bases, in square meters, in terms of π?

(A) 150π **(B) 170π ✓**

(C) 220π (D) 120π

1. Two bases: Type `2pi*5^2` : it returns 157.0796, which is 50π.

2. Curved side: Type `2pi*5*12` : it returns 376.9911, which is 120π.

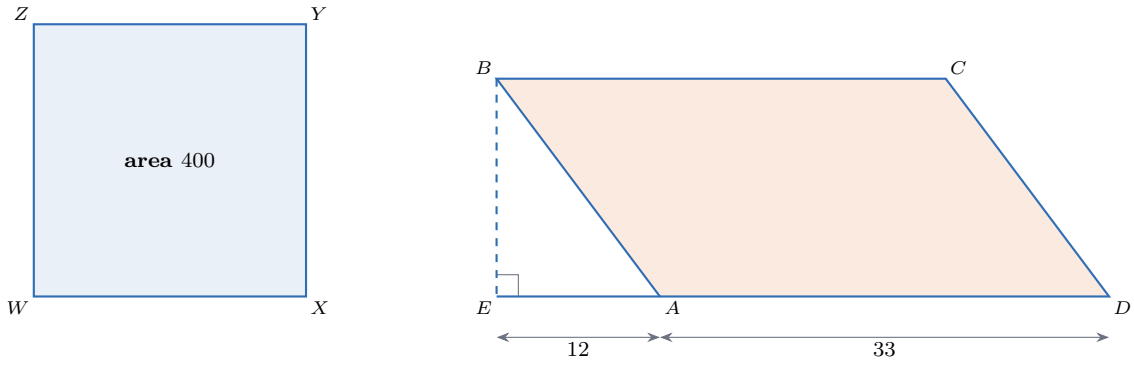
3. Total: Type `2pi*5^2+2pi*5*12` : it returns 534.0708.

4. Match the option: type `170pi` : it returns 534.0708, option (B).

Answer: (B)

7. HARD · DESMOS

In the figure, square $WXYZ$ has an area of 400 square units. Side AB of parallelogram $ABCD$ has the same length as a side of the square. Right triangle ABE, formed by dropping an altitude from vertex B to the extension of side AD, has hypotenuse AB and one leg $AE = 12$. If side AD of the parallelogram has length 33, what is the area of parallelogram $ABCD$?



1. The square's side: Type `sqrt(400)` : it returns 20, so $AB = 20$.

2. Leg BE of the right triangle: Type `sqrt(20^2-12^2)` : it returns 16.

3. Area of the parallelogram, base times height: Type `33*16` : it returns 528.

4. Trap check: using the slanted side AB as the height gives type `33*20` : it returns 660, which is wrong.

Answer: 528

8. HARD · DESMOS

A rectangular vegetable garden originally has length s meters and width $0.4s$ meters. A surrounding path is widened, which decreases the garden's length by $0.1s$ and its width by $0.1s$. What is the percent decrease in the garden's area?

1. Pick $s = 10$ so the numbers are easy. Original area: Type `10*0.4*10` : it returns 40.

2. New sides are $0.9s$ and $0.3s$. New area: Type `0.9*10*0.3*10` : it returns 27.

3. Percent decrease: Type `(40-27)/40*100` : it returns 32.5.

4. So the area decreases by 32.5%. Enter 32.5.

Answer: 32.5

9. HARD · DESMOS

A hemisphere is half of a sphere. If a hemisphere has a radius of 18 inches, which of the following is closest to the volume, in cubic inches, of the hemisphere?

(A) 24,400 (B) 2,400

(C) 6,100 **(D) 12,200 ✓**

1. Hemisphere volume: $V = \frac{2}{3}\pi r^3$ with $r = 18$. First the cube: type `18^3`: it returns 5832.

2. Type `2/3*pi*18^3`: it returns 12214.5122, so the volume is about 12,215 cubic inches.

3. The closest option is 12,200, option (D). The full sphere, $\frac{4}{3}\pi r^3$, would be about 24,429, which is the trap option.

Answer: (D)

10. HARD · DESMOS

A conical grain storage silo has a radius of 6 meters and a height of 10 meters. Which of the following is closest to the volume of the silo, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 376.8 ✓ (B) 452.2

(C) 565.2 (D) 282.6

1. Cone volume: $V = \frac{1}{3}\pi r^2 h$ with $\pi \approx 3.14$, $r = 6$, $h = 10$. Type `1/3*3.14*6^2*10`: it returns 376.8.

2. By hand: $6^2 = 36$, and $\frac{1}{3} \cdot 3.14 \cdot 36 \cdot 10 = 376.8$, option (A).

3. Check the traps: forgetting the $\frac{1}{3}$ gives 1130.4, and swapping r and h gives 628.

Answer: (A)

11. HARD · DESMOS

A spherical water tank has a radius of 5 meters. Which of the following is closest to the surface area of the tank, in square meters? (Use $\pi \approx 3.14$ and the formula $SA = 4\pi r^2$.)

(A) 261.7 **(B) 314.0 ✓**

(C) 392.5 (D) 235.5

1. Sphere surface area: $SA = 4\pi r^2$ with $\pi \approx 3.14$ and $r = 5$. First $r^2 = 25$.

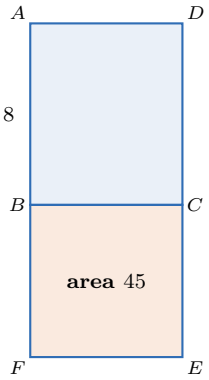
2. Type `4*3.14*5^2`: it returns 314.

3. That is option (B). A wrong formula gives a trap: for example $2\pi r^2$ gives 157.

Answer: (B)

12. HARD · DESMOS

Side AB of rectangle $ABCD$ measures 8 units. Side BC of the rectangle is shared with side BC of square $BCEF$. If the area of square $BCEF$ is 45 square units, what is the area of rectangle $ABCD$?



(A) 80 square units

(B) $12\sqrt{5}$ square units

(C) $24\sqrt{5}$ square units ✓

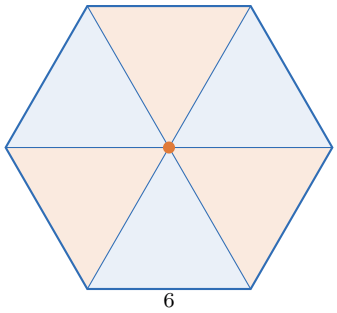
(D) $45 + 8\sqrt{5}$ square units

1. Side of the square: Type `sqrt(45)` : it returns 6.7082, which is $3\sqrt{5}$.
2. Rectangle area: Type `8*sqrt(45)` : it returns 53.6656.
3. Match the option: type `24sqrt(5)` : it returns 53.6656, option (C). The option type `45+8sqrt(5)` : it returns 62.8885 does not match.

Answer: (C)

13. HARD · DESMOS

A hexagon is formed by arranging 6 equilateral triangles, each of which has a vertex at the center of the hexagon. If the length of one side of a triangle is 6, the area of the hexagon is $k\sqrt{3}$ square units. What is the value of k?



1. Area of one equilateral triangle with side 6: Type `sqrt(3)/4*6^2` : it returns 15.5885, which is $9\sqrt{3}$.
2. Six triangles: Type `6*sqrt(3)/4*6^2` : it returns 93.5307.
3. Divide by $\sqrt{3}$ to find k: Type `6*6^2/4` : it returns 54.
4. Enter 54.

Answer: 54

14. HARD · DESMOS

Two similar rectangles have areas 630 and 2,520. The perimeter of the smaller rectangle is 210, and the perimeter of the larger is n. What is the value of n?

(A) 420 ✓ (B) 2,100

(C) 1,680 (D) 840

1. Area ratio: Type `2520/630` : it returns 4.
2. Scale factor: Type `sqrt(2520/630)` : it returns 2.
3. Perimeter of the larger rectangle: Type `2*210` : it returns 420.
4. Option (A).

Answer: (A)

15. HARD · DESMOS

A right square pyramid has a base area of 100 and a total surface area of $100 + 100\sqrt{5}$. What is the height of the pyramid?

1. Side of the base: Type `sqrt(100)` : it returns 10.
2. Slant height from the lateral area $100\sqrt{5}$: Type `100sqrt(5)/(4*1/2*10)` : it returns 11.1803, which is $5\sqrt{5}$.
3. Height: Type `sqrt((100sqrt(5)/(4*1/2*10))^2-5^2)` : it returns 10.
4. Enter 10.

Answer: 10

16. HARD · DESMOS

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

(A) 1500 (B) 180

(C) 300 **(D)** 500 ✓

1. Scale factor from the surface areas: Type `sqrt(400/144)` : it returns 1.6667, which is $\frac{5}{3}$.
2. Volume of the larger crate: Type `108*(sqrt(400/144))^3` : it returns 500.
3. Option (D).

Answer: (D)

17. HARD · DESMOS

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

1. Scale factor (small to large): Type `sqrt(208/325)` : it returns 0.8, which is $\frac{4}{5}$.
2. Volume of the smaller block: Type `375*(sqrt(208/325))^3` : it returns 192.
3. Enter 192.

Answer: 192

18. HARD · DESMOS

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 3 (B) 4

(C) 12 ✓ (D) 2

1. Volume: $\pi r^2 h = 48\pi$, so $h = \frac{48}{r^2}$. Surface area: $2\pi r^2 + 2\pi rh = 56\pi$, so $r^2 + rh = 28$. Substitute h: $r^2 + \frac{48}{r} = 28$.
2. In Desmos type `y=x^2+48/x` and `y=28` (letting x stand for r).
3. Tap the intersections with $x > 0$: (2, 28) and (4, 28). (The point with $x = -6$ is not allowed: a radius cannot be negative.)
4. Heights: type `48/2^2`: it returns 12 for $r = 2$, and type `48/4^2`: it returns 3 for $r = 4$. Only $r = 2$ has height greater than the diameter ($12 > 4$). The height is 12, option (C).

Answer: (C)

19. HARD · DESMOS

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A. What is the ratio of the volume of sphere B to the volume of sphere A?

(A) $\frac{9}{4}$ **(B) $\frac{27}{8}$ ✓**

(C) $\frac{5}{4}$ (D) $\frac{3}{2}$

1. Radius of A: Type `8pi/(2pi)`: it returns 4. Surface area of A: type `4pi*4^2`: it returns 201.0619.
2. Radius of B: Type `sqrt((4pi*4^2+80pi)/(4pi))`: it returns 6.
3. Volume ratio: Type `(6/4)^3`: it returns 3.375.
4. Match the option: type `27/8`: it returns 3.375, option (B).

Answer: (B)

20. HARD · DESMOS

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 240 (B) 130

(C) 360 **(D) 260 ✓**

1. Side of the base: Type `10sqrt(2)/sqrt(2)`: it returns 10.
2. Height from the volume: Type `3*400/10^2`: it returns 12.
3. Slant height: Type `sqrt(12^2+5^2)`: it returns 13.
4. Lateral area: Type `4*1/2*10*13`: it returns 260. Option (D).

Answer: (D)

21. HARD · DESMOS

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t?

1. Type `y=15*pi*(100-x^2)` and `y=765*pi` (letting x stand for the inner radius).
2. Tap the intersection with $0 < x < 10$: (7, 2403.3184). The inner radius is 7.
3. Thickness: Type `10-7`: it returns 3.
4. Enter 3.

Answer: 3

22. HARD · DESMOS

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

(A) 936 ✓ (B) 1160

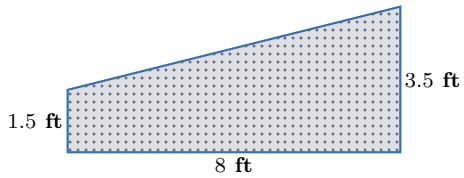
(C) 616 (D) 866

1. Half the base sides: Type `sqrt(13^2-12^2)`: it returns 5 and type `sqrt(20^2-12^2)`: it returns 16, so the base is 10 by 32.
2. Lateral area: Type `2*(1/2*32*13)+2*(1/2*10*20)`: it returns 616.
3. Total with the base: Type `2*(1/2*32*13)+2*(1/2*10*20)+10*32`: it returns 936.
4. Option (A).

Answer: (A)

23. HARD · DESMOS

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?



cross-section, perpendicular to the length of the bed

1. The soil's cross-section is a trapezoid with parallel sides 1.5 and 3.5 and width 8. Type `(1.5+3.5)/2*8`: it returns 20 square feet.
2. Volume = cross-section area × length. Type `20*12`: it returns 240 cubic feet.
3. Convert to cubic yards: 1 yard = 3 feet, so 1 cubic yard = $3^3 = 27$ cubic feet. Type `240/27*45`: it returns 400 dollars.
4. The cost is 400 dollars.

Answer: 400

24. HARD · DESMOS

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q?

(A) 150π **(B) 375π ✓**

(C) 600π (D) 60π

1. Surface area of P: Type `2pi*2^2+2pi*2*6` : it returns 100.531, which is 32π.
2. Scale factor: Type `sqrt(200pi/(32pi))` : it returns 2.5.
3. Volume of Q: Type `pi*(2*2.5)^2*(6*2.5)` : it returns 1178.0972.
4. Match the option: type `375pi` : it returns 1178.0972, option (B).

Answer: (B)

25. HARD · DESMOS

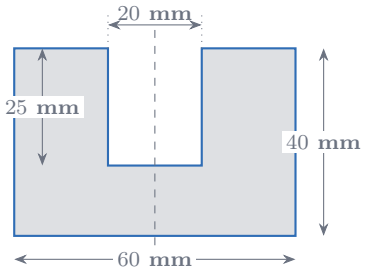
Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q?

1. Box P: surface area Type `2(3*4+3*6+4*6)` : it returns 108 and volume 72.
2. Scale factor: Type `(1125/(3*4*6))^(1/3)` : it returns 2.5.
3. Surface area of Q: Type `108*2.5^2` : it returns 675.
4. Enter 675.

Answer: 675

26. HARD · DESMOS

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)



cross-section through the axis

(A) 82,090 (B) 784

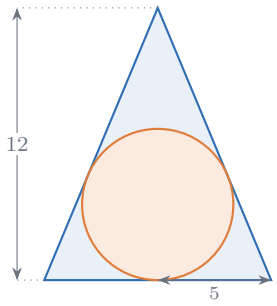
(C) 821 ✓ (D) 882

1. Convert to centimeters first (10 mm = 1 cm): the cylinder has radius 3 cm and height 4 cm, and the hole has radius 1 cm and depth 2.5 cm.
2. Whole cylinder: Type `pi*3^2*4` : it returns 113.0973 (cm^3). Hole: type `pi*1^2*2.5` : it returns 7.854 (cm^3).
3. Metal left: Type `pi*3^2*4-pi*1^2*2.5` : it returns 105.2434 (cm^3).
4. Mass = volume × density. Type `(pi*3^2*4-pi*1^2*2.5)*7.8` : it returns 820.8982, so about 821 grams, option (C). Ignoring the hole gives 882, the trap.

Answer: (C)

27. Hard · Desmos

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)



1. Slanted side of the axial triangle: Type `sqrt(5^2+12^2)` : it returns 13.
2. Radius of the inscribed circle, area over semiperimeter: Type `(1/2*10*12)/((13+13+10)/2)` : it returns 3.3333, which is $\frac{10}{3}$.
3. Fraction of the cone filled: Type `(4/3*pi*(10/3)^3)/(1/3*pi*5^2*12)` : it returns 0.4938.
4. Match the fraction: type `40/81` : it returns 0.4938. Enter $\frac{40}{81}$.

Answer: $\frac{40}{81}$

28. Hard · Desmos

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 67.5 ✓ (B) 202.5

(C) 7.5 (D) 22.5

1. Surface-area ratio: Type `14.4/1.6` : it returns 9.
2. Scale factor: Type `sqrt(9)` : it returns 3.
3. Volume of the copy: Type `2.5*3^3` : it returns 67.5.
4. Option (A).

Answer: (A)

29. Hard · Desmos

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

1. Volume ratio of the aquariums: $\frac{432}{250} = 1.728$. Volumes scale by k^3, so find k: Type `(432/250)^(1/3)` : it returns 1.2.
2. Glass area scales by k^2. Type `400*1.2^2` : it returns 576 square decimeters.
3. The larger aquarium has 576 square decimeters of glass. (Using k^3 instead would give 691.2.)

Answer: 576

30. Hard · Desmos

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and each of the four edges from the apex to a corner of the base has length 13 inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 260 (B) 120

(C) 340 **(D) 240 ✓**

1. Side of the base: Type `sqrt(100)` : it returns 10.
2. Height of one triangular face: Type `sqrt(13^2-5^2)` : it returns 12.
3. Four faces: Type `4*1/2*10*12` : it returns 240.
4. Option (D). Using 13 as the face height gives type `4*1/2*10*13` : it returns 260, the trap.

Answer: (D)

31. Hard · Desmos

Two similar rectangular prisms have volumes in the ratio $8 : 27$. The smaller prism has a surface area of 40. What is the surface area of the larger prism?

1. Scale factor from the volume ratio $27 : 8$: Type `(27/8)^(1/3)` : it returns 1.5.
2. Surface area ratio: Type `1.5^2` : it returns 2.25.
3. Larger surface area: Type `40*1.5^2` : it returns 90.
4. Enter 90.

Answer: 90

32. Hard · Desmos

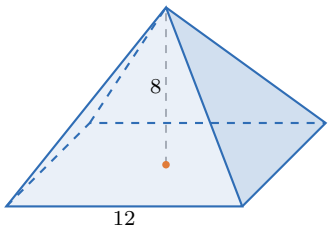
A rectangular tank has a base that is 40 centimeters by 30 centimeters. A solid cube with edge 12 centimeters is placed in the tank and sinks to the bottom, fully under water. By how many centimeters does the water level rise?

1. The water level rises because the cube pushes aside its own volume of water. Type `12^3` : it returns 1728 cubic centimeters.
2. That extra volume spreads over the base of the tank. Type `40*30` : it returns 1200 square centimeters.
3. Rise $= \dfrac{\text{volume}}{\text{base area}}$: Type `1728/1200` : it returns 1.44 centimeters.

Answer: 1.44

33. HARD · DESMOS

A square pyramid has a base with side length 12 and a height of 8. What is the lateral surface area of the pyramid (the four triangular faces, excluding the base)?



1. Slant height of a face: Type `sqrt(8^2+6^2)` : it returns 10.
2. One face: Type `1/2*12*10` : it returns 60.
3. Four faces: Type `4*60` : it returns 240.
4. Enter 240.

Answer: 240

34. HARD · DESMOS

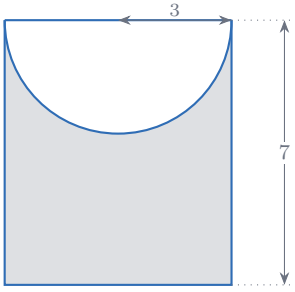
The radius of a sphere is increased by 20%. By what percent does the volume of the sphere increase? (Enter as a decimal.)

1. Increasing the radius by 20% multiplies it by 1.2. Volume depends on r^3, so the volume is multiplied by 1.2^3: Type `1.2^3` : it returns 1.728.
2. A multiplier of 1.728 means an increase of 72.8%: Type `(1.2^3-1)*100` : it returns 72.8.
3. Enter 72.8. (The trap $3 \times 20 = 60$ forgets that the change compounds.)

Answer: 72.8

35. HARD · DESMOS

A solid is formed by carving a hemispherical bowl of radius 3 out of the top of a right circular cylinder of radius 3 and height 7. The volume of the solid is $k\pi$. What is the value of k?



cross-section through the axis

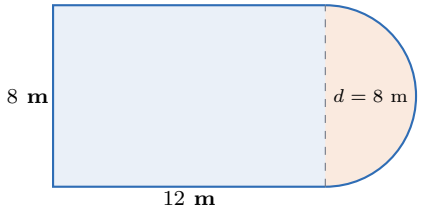
1. Cylinder: Type `pi*3^2*7` : it returns 197.9203.
2. Bowl (hemisphere): Type `2/3*pi*3^3` : it returns 56.5487.
3. Solid divided by π: Type `(pi*3^2*7-2/3*pi*3^3)/pi` : it returns 45.
4. Enter 45.

Answer: 45

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · DESMOS

A school courtyard is shaped like a 12-meter by 8-meter rectangle with a semicircular reading nook attached to one of the 8-meter sides, as shown. The semicircle has a diameter of 8 meters. What is the total area of the courtyard, in square meters?



(A) 104π

(B) $96 + 4\pi$

(C) $96 + 8\pi$ ✓

(D) $96 + 16\pi$

1. The courtyard is a rectangle plus a half circle. Type `12*8` : it returns 96 for the rectangle.
2. The half circle has radius $8 \div 2 = 4$. Type `1/2*pi*(8/2)^2` : it returns 25.1327, which is 8π.
3. Type `12*8+1/2*pi*(8/2)^2` : it returns 121.1327 square meters.
4. Type each option to match: type `96+8pi` : it returns 121.1327, option (C). The option type `96+4pi` : it returns 108.5664 does not match.

Answer: (C)

37. MEDIUM · DESMOS

A cylindrical water tank supplying a public fountain has a radius of 15 meters and a volume of 4500π cubic meters. What is the height of the tank, in meters?

(A) 15

(B) 20 ✓

(C) 30

(D) 10

1. Type `y=pi*15^2*x` and `y=4500*pi` (letting x stand for the height).
2. Tap the intersection: (20, 14137.1669). Its x-value is the height.
3. Type `4500/15^2` : it returns 20, the same height in meters.
4. Check: type `pi*15^2*20` : it returns 14137.1669 equals type `4500pi` : it returns 14137.1669. Option (B).

Answer: (B)

38. MEDIUM · DESMOS

A rectangular storage box has length 8 cm, width 5 cm, and height 4 cm. What is the total surface area of the box, in square centimeters?

(A) 92

(B) 144

(C) 160

(D) 184 ✓

1. Area of the three different faces: Type `8*5` : it returns 40, type `8*4` : it returns 32, type `5*4` : it returns 20.
2. Each face appears twice. Type `2(8*5+8*4+5*4)` : it returns 184.
3. So the total surface area is 184 square centimeters, option (D).

Answer: (D)

39. Medium · Desmos

A cube has a total surface area of 150 square centimeters. What is the volume of the cube, in cubic centimeters?

(A) 125 ✓ (B) 150

(C) 25 (D) 75

1. Type `y=6x^2` and `y=150` (letting x stand for the edge s).

2. Tap the intersection with $x > 0$: (5, 150). The edge is the x-value.

3. Type `5^3`: it returns 125 cubic centimeters.

4. Option (A).

Answer: (A)

40. Medium · Desmos

A rectangular water storage tank at a facility has a base measuring 8 feet by 5 feet. The tank is filled with water to a depth of 4.5 feet. Using a water density of 62.4 pounds per cubic foot, what is the weight of the water in the tank, in pounds?

1. Volume of water = base area times depth. Type `8*5*4.5`: it returns 180 cubic feet.

2. Weight = volume times density. Type `180*62.4`: it returns 11232 pounds.

3. Enter 11232 (no comma) as the answer.

Answer: 11232

41. Medium · Desmos

A cylindrical water tank has a base circumference of 44π meters and a height of 15 meters. The volume of the tank is $k\pi$ cubic meters. What is the value of k?

1. The circumference of the base is $2\pi r = 44\pi$, so $r = \dfrac{44\pi}{2\pi} = 22$ meters.

2. Volume of a cylinder: $V = \pi r^2 h = \pi(22)^2(15)$. Type `22^2*15`: it returns 7260, so $V = 7260\pi$.

3. The question writes the volume as $k\pi$, so $k = 7260$.

Answer: 7260

42. Medium · Desmos

A design team is producing rectangular flag replica banners of two sizes. Rectangle A is similar to rectangle B, and the side lengths of rectangle A are 3 times the corresponding side lengths of rectangle B. If the area of rectangle B is 24 square centimeters, what is the area, in square centimeters, of rectangle A?

1. Area scales by the square of the scale factor. Type `3^2`: it returns 9.

2. Type `24*3^2`: it returns 216 square centimeters.

3. Trap check: type `24*3`: it returns 72 would be the answer if area scaled like length.

4. Enter 216.

Answer: 216

43. Medium · Desmos

For a school engineering fair project, a team builds a right circular cone-shaped sand mound model with a volume of 48π cubic centimeters and a height of 9 centimeters. What is the radius, in centimeters, of the base of the cone?

1. Type `y=1/3*pi*x^2*9` and `y=48*pi` (letting x stand for the radius).

2. Tap the intersection with $x > 0$: (4, 150.7964). The radius is the x-value.

3. Check: Type `1/3*pi*4^2*9`: it returns 150.7964 and type `48pi`: it returns 150.7964 agree.

4. The radius is 4 centimeters.

Answer: 4

44. Medium · Desmos

A cylindrical water tank has a radius of 3 meters and a height of 8 meters. Which of the following is closest to the volume of the tank, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 188.4

(B) 226.1 ✓

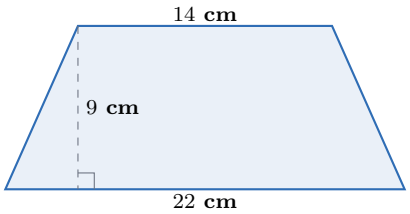
(C) 254.3

(D) 150.7

1. Cylinder: $V = \pi r^2 h$ with $\pi \approx 3.14$, $r = 3$, $h = 8$. Type `3.14*3^2*8`: it returns 226.08.
2. Round to one decimal place: 226.1, option (B).
3. Check the traps: forgetting to square r gives 75.4, and using the diameter 6 as the radius gives 904.3.
Answer: (B)

45. Medium · Desmos

A trapezoid has parallel sides of length 14 cm and 22 cm, and a height of 9 cm. What is the area of the trapezoid, in square centimeters?



(A) 144

(B) 153

(C) 162 ✓

(D) 171

1. Add the parallel sides and halve: Type `(14+22)/2`: it returns 18.
2. Multiply by the height: Type `(14+22)/2*9`: it returns 162 square centimeters.
3. Option (C).
Answer: (C)

46. Medium · Desmos

For a particular square, the perimeter is numerically equal to the area. What is the perimeter of the square?

1. Type `y=x^2` and `y=4x` (area and perimeter of a square with side x).
2. Tap the intersection with $x > 0$: (4, 16). The side is 4.
3. Perimeter: Type `4*4`: it returns 16.
4. Enter 16.
Answer: 16

47. Medium · Desmos

A circular coaster has a diameter of 10 cm and a height (thickness) of 3 mm. If 8 coasters are stacked on top of each other, what is the volume of the stack?

(A) 60π cm^3 ✓ (B) 75π cm^3

(C) 120π cm^3 (D) 150π cm^3

1. Radius: Type `10/2`: it returns 5. Thickness in centimeters: type `3/10`: it returns 0.3.

2. All 8 coasters together: Type `8*pi*5^2*0.3`: it returns 188.4956.

3. Match the option: type `60pi`: it returns 188.4956, option (A).

Answer: (A)

48. Medium · Desmos

If the surface area of a cube is 96 square centimeters, what is the volume of the cube, in cubic centimeters?

1. Type `y=6x^2` and `y=96` (letting x stand for the edge).

2. Tap the intersection with $x > 0$: (4, 96).

3. Volume: Type `4^3`: it returns 64 cubic centimeters.

4. Enter 64.

Answer: 64

49. Medium · Desmos

Two similar rectangular prisms have volumes 972 and 36. An edge of the larger prism has length 12. What is the length of the corresponding edge of the smaller prism?

1. Volume ratio: Type `972/36`: it returns 27.

2. Scale factor: Type `(972/36)^(1/3)`: it returns 3.

3. Smaller edge: Type `12/3`: it returns 4.

4. Enter 4.

Answer: 4

50. Medium · Desmos

The edges of cubes A and B are in the ratio $\frac{3}{88}$ (edge of A to edge of B). What is the ratio of the surface area of B to the surface area of A?

(A) $\frac{9}{7{,}744}$ (B) $\frac{3}{88}$

(C) $\frac{88}{3}$ **(D) $\frac{7{,}744}{9}$ ✓**

1. Surface areas scale by the square of the edge ratio $B : A = \frac{88}{3}$. Type `(88/3)^2`: it returns 860.4444.

2. Match the fraction option: type `7744/9`: it returns 860.4444.

3. The reversed fraction gives type `(3/88)^2`: it returns 0.0012, which is the trap. Option (D).

Answer: (D)

51. Medium · Desmos

A right circular cylinder has volume 90π and height 10. The lateral surface area (excluding the bases) of the cylinder is $k\pi$. What is the value of k?

1. Type `y=pi*x^2*10` and `y=90*pi` (letting x stand for the radius).

2. Tap the intersection with $x > 0$: (3, 282.7433). The radius is 3.

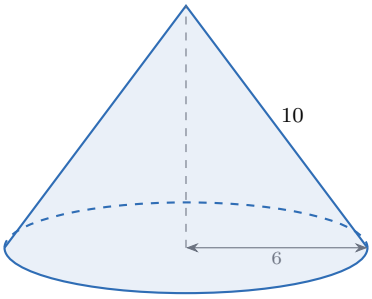
3. Lateral area divided by π: Type `2*3*10`: it returns 60.

4. Enter 60.

Answer: 60

52. Medium · Desmos

A right circular cone has base radius 6 and slant height 10. The volume of the cone is $k\pi$. What is the value of k?



1. Height from the right triangle: Type `sqrt(10^2-6^2)`: it returns 8.

2. Volume divided by π: Type `1/3*6^2*8`: it returns 96.

3. Enter 96.

Answer: 96

53. Medium · Desmos

A sphere has surface area 144π. The volume of the sphere is $k\pi$. What is the value of k?

1. Type `y=4*pi*x^2` and `y=144*pi` (letting x stand for the radius).

2. Tap the intersection with $x > 0$: (6, 452.3893). The radius is 6.

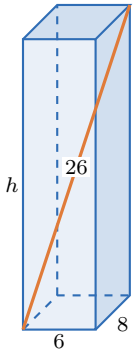
3. Volume divided by π: Type `4/3*6^3`: it returns 288.

4. Enter 288.

Answer: 288

54. Medium · Desmos

A rectangular box has a base that measures 6 by 8. The distance from one corner of the box to the opposite corner (the space diagonal) is 26. What is the height of the box?



1. Type `y=6^2+8^2+x^2` and `y=26^2` (letting x stand for the height).

2. Tap the intersection with $x > 0$: (24, 676). The height is 24.

3. Check: Type `sqrt(26^2-6^2-8^2)` : it returns 24.

4. Enter 24.

Answer: 24

55. Medium · Desmos

The space diagonal of a cube (the segment joining opposite vertices) has length $9\sqrt{3}$. What is the volume of the cube?

1. Edge from the space diagonal $e\sqrt{3} = 9\sqrt{3}$: Type `9sqrt(3)/sqrt(3)` : it returns 9.

2. Volume: Type `9^3` : it returns 729.

3. Trap check: type `9^2` : it returns 81 is a face area, not the volume.

4. Enter 729.

Answer: 729

56. Medium · Desmos

A box has interior dimensions 10 cm by 12 cm by 16 cm. Cubes with edge length 2 cm are packed into the box with no gaps. How many cubes fit?

1. Cubes along each edge: Type `10/2` : it returns 5, type `12/2` : it returns 6, type `16/2` : it returns 8.

2. Total: Type `5*6*8` : it returns 240.

3. Check with volumes: Type `10*12*16/2^3` : it returns 240.

4. Enter 240.

Answer: 240

57. Medium · Desmos

A cone and a cylinder each have radius 5 and height 12. The cone is filled with water and the water is poured into the cylinder, repeatedly, until the cylinder is full. How many full cones of water does it take to fill the cylinder?

1. Cylinder: Type `pi*5^2*12` : it returns 942.4778.

2. Cone: Type `1/3*pi*5^2*12` : it returns 314.1593.

3. Number of cones: Type `(pi*5^2*12)/(1/3*pi*5^2*12)` : it returns 3.

4. Enter 3.

Answer: 3